Ruiyao Chen

Data Scientist · Earth Observation, Machine Learning & Uncertainty

Princeton, NJ (Eastern Time) chenruiyao@gmail.com www.linkedin.com/in/ruiyao-chen ruiyao-chen.github.io

U.S. permanent resident

SUMMARY

Ph.D. remote-sensing data scientist with 10+ years turning satellite and environmental data into validated products for NASA missions (TROPICS, GPM, TRMM, PolSIR). I specialize in the part most models skip: defining what "accurate" means for a product, designing the validation that proves it, and quantifying uncertainty for every prediction. I combine radiative-transfer physics with machine learning in Python, and I have taken models from research prototype to production code running in NASA's operational processing.

SKILLS

Machine learning & statistics	XGBoost, random forests, PyTorch (CNN, MLP), conformal prediction, quantile regression, k-d tree / instance-based retrieval, Monte Carlo UQ, spatial cross-validation, feature engineering
Python & data	Python, NumPy, SciPy, pandas, xarray, GeoPandas, rasterio, scikit-learn, pytest, SQL, Git, Linux, HPC
Earth observation	Google Earth Engine, GEDI, Sentinel-2, Landsat 8/9, passive microwave, GOES-16/17, ERA5, GEOS-5, CRTM, RTTOV, ARTS
Also	Fortran, MATLAB, R, C/C++, technical writing, independent-review defense, mentoring

EXPERIENCE

Research Scientist

2023 – present

Vanderbilt University, Earth & Environmental Sciences · Nashville, TN / Princeton, NJ

- Own NASA TROPICS' operational precipitation ice-water-path retrieval end to end: proposed the algorithm, built and validated it, wrote the production code and documentation, delivered it to NASA for operational processing, and improved later versions from user feedback.
- Found brightness-temperature calibration drift between time periods in TROPICS Level-1 data that was propagating into retrievals; built per-period bias corrections from clear-sky radiative-transfer simulations, confirmed with the mission calibration team.
- Built per-pixel uncertainty estimates for the ice retrieval and documented what they capture versus total error, so users read confidence correctly.
- Built pre-launch simulated observations for NASA PolSIR (CRTM radiative transfer from ERA5 with per-channel noise, matched to ERA5, GEOS-5 and GOES-16/17), giving algorithm teams a known-truth testbed years before launch.

Postdoctoral Researcher

2018 – 2023

Vanderbilt University, Earth & Environmental Sciences · Nashville, TN

- Co-PI and technical lead on a NASA GPM grant characterizing uncertainty in the multi-satellite precipitation record; built Monte Carlo uncertainty-quantification frameworks for precipitation and cloud retrievals.
- Developed deep-learning (CNN with attention, MLP) and random-forest models for hydrometeor classification and rain-rate estimation from passive microwave data, grounded in radiative-transfer physics.
- Built Python, Fortran and SQL pipelines to aggregate, clean and validate multi-terabyte satellite datasets.

Research Scientist

2018

Environmental Model and Data Optima · Laurel, MD

- Designed gradient-boosted (XGBoost) models to cross-calibrate a network of satellite microwave sensors, reducing systematic bias by 11% across precipitation, weather and climate products.
- Developed cloud, haze and low-cloud detection algorithms from multispectral and microwave observations.

Graduate Research Assistant

2012 – 2018

Central Florida Remote Sensing Lab, University of Central Florida · Orlando, FL

- Root-caused two hidden on-orbit anomalies in NASA's 17-year TRMM Microwave Imager record and built physics-based corrections adopted in the TMI legacy (1B11 V8) reprocessing.
- With the NASA GPM inter-calibration working group, designed a three-way cross-calibration of GMI, TMI and WindSat, producing a consistent multi-decadal record (TMI–WindSat stable to 0.1 K over 5+ years).

EDUCATION

Ph.D., Electrical Engineering

2018

University of Central Florida · Stochastic modeling, radiative transfer and microwave remote sensing. Dissertation: a consistent oceanic multi-decadal inter-calibrated TMI–GMI data record.

M.S., Electrical Engineering

2014

University of Central Florida

B.Eng., Electronics & Information Engineering (Honors)

2012

Wuhan University of Science and Technology

PUBLICATIONS

2026 **TROPICS precipitation ice water path retrieval: a probabilistic k-d tree approach with uncertainty quantification**
R. Chen, R. Bennartz, R. V. Leslie, W. J. Blackwell. *IEEE IGARSS 2026 (journal version under review at IEEE Transactions on Geoscience and Remote Sensing)*

2023 **A deep-learning scheme for hydrometeor type classification using passive microwave observations**
R. Chen, R. Bennartz. *Remote Sensing, 15(10), 2670*

2021 **Rainfall algorithms using oceanic satellite observations from MWHS-2**
R. Chen, R. Bennartz. *Advances in Atmospheric Sciences, 38(8), 1367–1378*

2020 **Sensitivity of 89–190-GHz microwave observations to ice particle scattering**
R. Chen, R. Bennartz. *Journal of Applied Meteorology and Climatology, 59(7), 1195–1215*

2020 **Creating a consistent multi-decadal oceanic TRMM–GPM brightness temperature record with estimated calibration uncertainty**
R. Chen, W. L. Jones. *Climate, 8(2), 31*

2018 **Emissive reflector correction in the legacy version 1B11 V8 (GPM05) brightness temperature of the TRMM Microwave Imager**
F. Alquaied, **R. Chen**, W. L. Jones. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*

2018 **Hot load temperature correction for TRMM Microwave Imager in the legacy brightness temperature**
F. Alquaied, **R. Chen**, W. L. Jones. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*

2017 **Creating a multidecadal ocean microwave brightness dataset: three-way intersatellite radiometric calibration among GMI, TMI, and WindSat**
R. Chen, H. Ebrahimi, W. L. Jones. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*

2017 **Calibration of millimeter wave sounder radiometers on polar orbiting satellites**
H. Ebrahimi, **R. Chen**, W. L. Jones. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*

2017 **Assessing radiometric stability of the 17-plus-year TRMM Microwave Imager 1B11 Version-8 (GPM05) brightness temperature product**
R. Chen, F. Alquaied, W. L. Jones. *Climate, 5(4), 92*

2016 **Intercalibration of the GPM microwave radiometer constellation**
W. Berg, S. Bilanow, **R. Chen**, S. Datta, D. Draper, H. Ebrahimi, S. Farrar, et al. *Journal of Atmospheric and Oceanic Technology, 33(12), 2639–2654*

GRANTS & HONORS

- Co-PI, Characterizing Uncertainty of Global Precipitation Measurement (GPM), NASA, 2019–2020
- NASA Agency Group Achievement Award, 2015
- Graduate Dean's Dissertation Completion Fellowship, UCF, 2018
- NSF L.E.A.R.N. Program Mentor Scholarship, 2015
- Machine Learning certificate, Stanford University (online), 2018

SERVICE

- Panelist, NASA ROSES proposal review
- Reviewer for IEEE TGRS, JSTARS and GRSL, and for AMS and AGU journals